

ANURAG REDDY ADMA

Software Engineer | AI & ML Specialist | Full-Stack & Embedded Systems

Hyderabad, Telangana, India | +91 91822 36284 | admaanuragreddy@gmail.com | LinkedIn: linkedin.com/in/adma-anurag-reddy | GitHub: github.com/anuragreddy

PROFESSIONAL SUMMARY

Innovative Software Engineer and B.Tech CSE (AI & ML) student with proven industry experience as an Automation Intern at T-Works and Web Development Intern at eKincare. Proven track record building enterprise hackathon management platforms (LRC 2026), industrial embedded HMI smart interfaces (ITC Automated Roti Maker), and deep learning computer vision diagnostic models. Skilled across Python, React, JavaScript/TypeScript, Node.js, MySQL, TensorFlow, and DWIN DGUS embedded architectures.

WORK EXPERIENCE

T-Works | Automation Engineering Intern

Hyderabad, India

- **Embedded HMI & Industrial Automation:** Engineered industrial Human-Machine Interface (HMI) smart display solutions for automated appliances (ITC Automated Roti Maker) using DWIN DGUS / T5L hardware, custom touch calibration, and UART protocols.
- **Custom Automation Toolchain:** Developed custom Python automation toolchains for video-to-frame extraction, palette quantization, memory mapping, and ICL binary graphic asset compilation.
- **Hackathon Platform Architecture:** Contributed to the LRC 2026 Hackathon Management Platform, engineering database schemas (49+ tables), role-based evaluation workflows, and arena scoring modules.

CodeOrbit Tech | Software Intern

Worked as a Web Development Intern at CodeOrbit, gained hands-on experience in designing, developing, and maintaining web applications. Contributing to frontend and backend development while improving my skills in modern web technologies and real-world software development practices.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript (ES6+), TypeScript, C, SQL, Java (Core Concepts)

Frontend & Web: React.js, Vite, HTML5, CSS3, Tailwind CSS

Backend & APIs: Node.js, Express.js, RESTful APIs, JWT Authentication

AI & Machine Learning: Convolutional Neural Networks (CNNs), Deep Learning, TensorFlow, Keras, OpenCV, CLAHE, Scikit-Learn

Databases & Storage: MySQL, SQLite, Relational Schema Design (49+ Tables), Parameterized Queries, ACID Transactions

Embedded Systems & IoT: DWIN DGUS Displays, T5L ASIC Architecture, Serial/UART Protocols, Video-to-ICL Compilation, Touch Calibration

Tools & Workflows: Git, GitHub, Postman, VS Code, Linux/Windows CLI, Agile/Scrum, LeetCode (Problem Solving & DSA)

FEATURED PROJECTS

Lam Research Challenge Hackathon (LRC 2026) Platform | *React, Node.js, Express, MySQL (49+ Tables), REST APIs, RBAC* | URL: <https://lrc2026.tworks.in/>

- **Multi-Tier Architecture & Workflows:** Architected an end-to-end competition platform backing 49+ tables for T-Works Admins, Evaluators, Observers, and Jury Panels with real-time arena evaluation scoring and expense tracking.

ITC Automated Industrial Roti Maker — DWIN DGUS Smart Embedded HMI | *Python, DWIN DGUS Displays, T5L ASIC, UART Serial, C*

- **HMI Automation & Toolchain:** Engineered industrial smart display interfaces on DWIN DGUS / T5L hardware. Developed a custom Python video-to-frame compilation toolchain synchronized with motor controllers over UART serial.

Brain Stroke Prediction & Classification via Deep Learning | *Python, TensorFlow, Keras, OpenCV (CLAHE), Scikit-learn, CNN*

- **Medical Vision Diagnostic Engine:** Constructed an end-to-end medical image diagnostic pipeline deploying deep CNNs on brain CT scans with CLAHE contrast enhancement, skull-stripping artifact removal, and ROC-AUC evaluation.

Institutional Faculty Management System | *JavaScript (ES6+), Node.js, Express.js, MySQL, REST APIs, JWT Auth*

- **Academic Resource & Workload Administration:** Engineered a full-stack institutional faculty portal streamlining academic scheduling, workload allocation, faculty performance credentials, and attendance tracking with role-based access control.

EDUCATION

Malla Reddy College of Engineering — Secunderabad, Telangana

2

2023 – Expected 2027

Bachelor of Technology (B.Tech) in Computer Science & Engineering (AI & ML) | Coursework: DSA, DBMS, Operating Systems, Machine Learning, Deep Learning

Sri Chaitanya Junior College—Narsingi ,Telangana

2

2021 – 2023

Intermediate (MPC - Mathematics, Physics, Chemistry) | **Score: 92%**

GMR CV—Shamshabad,Telangana

2020-2021

SCHOOLING(CBSE)

CERTIFICATIONS & KEY HIGHLIGHTS

- **Industry Certification:** Virtual Reality (VR) & Augmented Reality (AR) Certification — ARK Infosolutions (March 2024)
- **Algorithmic Problem Solving:** Solved 100+ algorithmic and data structure problems on LeetCode covering Arrays, Dynamic Programming, and Graphs.
- **Languages:** Languages: English (Professional), Telugu (Native), Hindi (Professional).